CORE PURE / DIFFERENTIAL EQUATIONS

Modelling with 1st Order Differential Equations

Questions by topic 14 questions · 225 marks

Questions

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1. A particle B, of mass $2\,\text{kg}$, moves in a straight line and has velocity $v\,\text{m}\,\text{s}^{-1}$.

At time t seconds, where $t \geq 0$, a variable force of $\dfrac{12t - 4v}{1 + t}\,\text{N}$ acts on B.

There are no other forces acting on B. Initially, when $t = 0$, B has velocity $-5\,\text{m}\,\text{s}^{-1}$.

The motion of B can be modelled by the differential equation

$$\frac{\mathrm{d}v}{\mathrm{d}t} + P(t)v = Q(t)$$

where $P(t)$ and $Q(t)$ are functions of t.

(a) Find the functions $P(t)$ and $Q(t)$ **[2]**

(b) Using an integrating factor, determine the first time at which B is stationary according to the model **[8]**

2. A mixing tank initially contains 600 litres of pure water.

Brine flows into the tank at a constant rate of 20 litres per hour. Each litre of brine contains 3 grams of dissolved salt.

The well-stirred mixture is pumped out of the tank at a constant rate of 30 litres per hour.

It is assumed that the salt is instantly and uniformly mixed throughout the liquid.

Given that there are x grams of salt in the tank after t hours,

(a) show that, while the tank still contains liquid, the situation can be modelled by

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 60 - \frac{3x}{60 - t} \qquad \textbf{[4]}$$

(b) Hence find the number of grams of salt in the tank after 30 hours **[6]**

(c) State one limitation of this model and explain briefly how the model could be refined **[1]**

3. A horticultural mixing tank initially contains 600 litres of pure water.

A fertiliser solution flows into the tank at a constant rate of 6 litres per minute. The incoming solution contains 2 grams of fertiliser in every litre of solution.

At the same time, liquid is pumped out of the well-stirred tank at a constant rate of 9 litres per minute.

It is assumed that the fertiliser instantly dissolves uniformly throughout the tank.

Given that there are x grams of fertiliser in the tank after t minutes,

(a) Show that, while the tank still contains liquid, the situation can be modelled by the differential equation

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 12 - \frac{3x}{200 - t}$$ **[4]**

(b) Hence determine after how many minutes the concentration of fertiliser in the tank first reaches 1.5 grams per litre. **[8]**

(c) Explain briefly one way in which the model could be refined. **[1]**

4. A processing tank initially contains 15 litres of water in which 6 mg of dye is dissolved.

A dye solution enters the tank at a constant rate of 200 ml per minute. The concentration of dye in this incoming solution, at time t minutes after the flow begins, is modelled by $(4 + e^{-0.1t})$ mg per litre.

At the same time, pure water enters through a second pipe at a constant rate of 100 ml per minute. The liquid in the tank is continually mixed and leaves the tank at a constant rate of 300 ml per minute.

Let m be the amount of dye, in milligrams, in the tank at time t minutes after the flow begins.

(a) (i) Write down the volume of liquid in the tank at time t
(ii) Hence show that m satisfies the differential equation

$$\frac{\mathrm{d}m}{\mathrm{d}t} = 0.8 + 0.2e^{-0.1t} - \frac{m}{50}$$ **[4]**

(b) By solving the differential equation, find an expression for m in terms of t **[5]**

After 30 minutes, the concentration of dye in the tank was measured as 1.52 mg per litre.

(c) Assess the model in light of this information, giving a reason for your answer **[2]**

5. A particle B, of mass 2 kg, moves in a straight line and has velocity $v\,\mathrm{m\,s^{-1}}$.

At time t seconds, where $t \geq 0$, a variable force of $-2t(v+3)\,\mathrm{N}$ acts on B. There are no other forces acting on B. Initially, when $t = 0$, B has velocity $5\,\mathrm{m\,s^{-1}}$.

The motion of B can be modelled by the differential equation

$$\frac{\mathrm{d}v}{\mathrm{d}t} + P(t)v = Q(t)$$

where $P(t)$ and $Q(t)$ are functions of t.

(a) Find the functions $P(t)$ and $Q(t)$ **[2]**

(b) Using an integrating factor, determine the first time at which B is stationary according to the model **[7]**

6. A brief flushing cycle in an industrial pipe is to be modelled. The flow rate of liquid, R litres per second, at time t seconds is measured from the instant the valve is opened. The cycle starts when $t = 0$ and there is initially no flow through the pipe. When $t = 1$, the flow rate is measured as $\pi e^{-1} \sin(1)$ litres per second.

While the cycle is occurring, R is modelled by the differential equation

$$\frac{\mathrm{d}R}{\mathrm{d}t} + (1 - \cot t)R = \frac{e^{-t} \sin t}{1 + t^2}$$

(a) By using an integrating factor show that, according to the model, while the cycle is occurring,

$$R = e^{-t} \sin t \left(\tan^{-1} t + \frac{3\pi}{4} \right)$$

[6]

The cycle ends when there is again no flow through the pipe.

(b) Determine the length of time that the cycle lasts according to the model **[2]**

(c) Determine, according to the model, the rate of increase of the flow rate at the start of the cycle. Give your answer in exact form **[3]**

7. A short filling pulse in a buffer tank is to be modelled. The volume of liquid in the tank at time t seconds is V litres. The pulse starts when $t = 0$, and initially the tank is empty. After 1 second the volume is measured to be 4 litres. While the pulse is occurring, V is modelled by the differential equation

$$(3t - t^2)\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{(3t - t^2)^2}{1 + (t-1)^2} + (3 - 2t)V$$

(a) By using an integrating factor show that, according to the model, while the pulse is occurring,

$$V = (3t - t^2)\left(\tan^{-1}(t-1) + 2\right)$$ **[6]**

The pulse ends when the tank is again empty.

(b) Determine the length of time that the pulse lasts according to the model **[2]**

(c) Determine, according to the model, the rate of increase of the volume at the start of the pulse. Give your answer in an exact form **[3]**

8. The concentration C, in suitable units, of a pollutant in a reservoir at time t hours after a filtration system is switched on is to be modelled by a differential equation. Initially, the concentration is A, and its long-term value is 0.

(a) One simple model is to assume that the rate of decrease of concentration is directly proportional to C.

(i) Formulate a differential equation for this model **[1]**

(ii) Verify that $C = Ae^{-kt}$, where k is a positive constant, satisfies this differential equation, the initial condition and the long-term condition **[3]**

An alternative model uses the differential equation

$$\frac{\mathrm{d}C}{\mathrm{d}t} + \frac{2t}{1+t^2}C = R(t)$$

where $R(t)$ is a function of t.

(b) Find the integrating factor for this differential equation, showing that it can be written in the form

$$1 + t^2$$ **[2]**

(c) Suppose that $R(t) = 0$.

(i) Show that

$$C = \frac{A}{1+t^2}$$ **[4]**

(ii) Find the time predicted by this model for the concentration to fall to half its initial value. Give your answer correct to the nearest minute **[2]**

(d) Now suppose that

$$R(t) = \frac{te^{-t}}{1+t^2}$$

Show that

$$C = \frac{A + 1 - (1+t)e^{-t}}{1+t^2}$$ **[6]**

It is found that the initial concentration is 5, and the concentration falls to half this value after 63 minutes.

(e) Determine which of the models proposed in parts (c) and (d) is more consistent with these data **[2]**

9. A laboratory mixing vessel initially contains 4 litres of solution in which 4 mg of dye is dissolved.

A wash solution enters the vessel at a constant rate of 300 ml per minute.

At the same time, well-mixed liquid leaves the vessel at a constant rate of 200 ml per minute.

The concentration of dye in the liquid entering the vessel, at time t minutes after pumping begins, is modelled by $2e^{-0.1t}$ mg per litre.

Let m be the amount of dye, in milligrams, in the vessel at time t minutes.

(a) Show that the volume of liquid in the vessel is $(4 + 0.1t)$ litres, and hence show that m satisfies

$$\frac{\mathrm{d}m}{\mathrm{d}t} = 0.6e^{-0.1t} - \frac{2m}{40 + t}$$ **[4]**

(b) Solve this differential equation, given that $m = 4$ when $t = 0$, to find an expression for m in terms of t **[9]**

(c) After 20 minutes, the concentration of dye in the vessel was measured as 0.82 mg per litre. Assess the model in light of this information, giving a reason for your answer **[2]**

10. A small rescue boat moves in a straight line on calm water. It starts from rest, is driven by its engine for 5 seconds and then, for the next 5 seconds, the engine is cut and a drogue is deployed. The boat comes to rest at the end of the 10 seconds. The total mass of the boat and crew is m kg, and at time t seconds, for $0 \leq t \leq 10$, the boat's velocity is $v\,\mathrm{m\,s^{-1}}$.
A resistance to motion, modelled by a force of magnitude $0.2mv$ N, acts on the boat throughout the 10 seconds.

(a) Explain why modelling the resistance to motion in this way is likely to be more realistic than assuming this force is constant **[1]**

(b) During the drogue phase of the motion, for $5 \leq t \leq 10$, the drogue exerts an additional constant resistance force of magnitude m N and the engine provides no driving force. Show that, for $5 \leq t \leq 10$,

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.2v = -1$$

[1]

(c) (i) Solve the differential equation in part (b) **[5]**
(ii) Hence find the velocity of the boat when $t = 5$ **[1]**

(d) During the powered phase ($0 \leq t \leq 5$), the propeller provides a driving force of magnitude directly proportional to t^2. Show that, for $0 \leq t \leq 5$,

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.2v = \lambda t^2$$

[1]

where λ is a positive constant

(e) (i) Show by integration that, for $0 \leq t \leq 5$,

$$v = 5\lambda\left(t^2 - 10t + 50 - 50e^{-t/5}\right)$$

[8]

(ii) Hence find λ **[2]**

(f) Find the total distance, to the nearest metre, travelled by the boat during the motion **[6]**

11. A winch raises a lift cage vertically. The cage starts from rest and comes to rest again 8 seconds later. The combined mass of the cage and its load is m kg, and at time t seconds, for $0 \leq t \leq 8$, the cage has upward velocity $v\,\mathrm{m\,s^{-1}}$.
A resistance to the motion, modelled by a force of magnitude $0.25mv$ N, acts on the cage throughout the 8 seconds.
In this question, take $g = 9.8\,\mathrm{m\,s^{-2}}$.

(a) Explain why modelling the resistance to the motion in this way is likely to be more realistic than assuming this force is constant **[1]**

(b) During the final stage of the motion, for $4 \leq t \leq 8$, the winch provides a constant upward tension of magnitude $0.75mg$ N. Show that, for $4 \leq t \leq 8$,

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.25v = -\frac{1}{4}g$$

[1]

(c) (i) Solve the differential equation in part (b) **[5]**
(ii) Hence find the velocity of the cage when $t = 4$ **[1]**

(d) During the first stage of the motion, for $0 \leq t \leq 4$, the winch provides an upward tension of magnitude $m(g + \lambda t)$ N, where λ is a positive constant. Show that, for $0 \leq t \leq 4$,

$$\frac{\mathrm{d}v}{\mathrm{d}t} + 0.25v = \lambda t$$

[1]

(e) (i) Show by integration that, for $0 \leq t \leq 4$,

$$v = 4\lambda\left(t - 4 + 4e^{-t/4}\right)$$

[6]

(ii) Hence find λ **[2]**

(f) Find the total distance, to the nearest metre, through which the cage rises during the motion **[6]**

12. A mixing tank initially contains 2 litres of solution in which 4 grams of dye are dissolved.
A dye solution containing 10 grams of dye per litre is pumped into the tank at a constant rate of r litres per minute and mixes instantly and uniformly with the liquid already in the tank.
At the same time, liquid is removed from the tank at a constant rate of s litres per minute.
After t minutes, the tank contains m grams of dye.

(a) (i) Show that the concentration of dye in the tank after t minutes is

$$\frac{m}{2+(r-s)t}$$

grams per litre. **[2]**

(ii) Hence show that

$$\frac{\mathrm{d}m}{\mathrm{d}t} + \frac{sm}{2+(r-s)t} = 10r$$ **[2]**

(b) First, consider the case where $s = r$.

(i) Solve the differential equation to find m in terms of r and t. **[5]**

(ii) Given that after 2 minutes the concentration of dye in the tank is 8 grams per litre, find r. **[2]**

(c) Now consider the case where $s = 2r$.

(i) Explain why the differential equation in part (a)(ii) is no longer valid for $t \geq \frac{2}{r}$. **[1]**

(ii) Find the maximum mass of dye in the tank. **[10]**

13. A drinks manufacturer uses a mixing tank that initially contains 4 litres of water. A flavouring liquid, which is 60% fruit concentrate by volume, is pumped into the tank at a constant rate of p litres per minute. At the same time, the contents of the tank are thoroughly mixed and drawn off at a constant rate of q litres per minute.

After t minutes, let x litres be the amount of fruit concentrate in the tank. You may assume that mixing is instantaneous and uniform.

(a) (i) Show that the proportion of fruit concentrate in the tank after t minutes is

$$\frac{x}{4+(p-q)t}$$ **[2]**

(ii) Hence show that

$$\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{qx}{4+(p-q)t} = \frac{3p}{5}$$ **[2]**

(b) Now consider the case where $q = p$.

(i) Solve the differential equation to find x in terms of p and t. **[5]**

(ii) Given that after 4 minutes the liquid in the tank is 30% fruit concentrate, find the value of p. **[2]**

(c) Now consider the case where $q = 2p$.

(i) Explain why the differential equation in part (a)(ii) is no longer valid for $t \geq \frac{4}{p}$. **[1]**

(ii) Find the maximum amount of fruit concentrate in the tank before the tank empties. **[10]**

14. The concentration C, in suitable units, of a drug in a patient's bloodstream at time t hours after an intravenous infusion begins is to be modelled. Initially, the concentration is zero, and its long-term value is M.

(a) One simple model assumes that the rate of change of concentration is directly proportional to $M - C$.

(i) Formulate a differential equation for this model. **[1]**

(ii) Verify that $C = M(1 - e^{-kt})$, where k is a positive constant, satisfies this differential equation, the initial condition and the long-term condition. **[3]**

An alternative model uses, for $t > 0$, the differential equation

$$\frac{\mathrm{d}C}{\mathrm{d}t} - \frac{t+2}{t(1+t+t^2)}C = S(t)$$

where $S(t)$ is a function of t.

(b) Find the integrating factor for this differential equation, showing that it can be written in the form

$$\frac{1+t+t^2}{t^2}$$

[5]

(c) Suppose that $S(t) = 0$.

(i) Show that

$$C = \frac{Mt^2}{1+t+t^2}$$

[4]

(ii) Find the time predicted by this model for the concentration to reach half its long-term value. Give your answer correct to the nearest minute. **[2]**

(d) Now suppose that

$$S(t) = \frac{3t^2e^{-t}}{1+t+t^2}$$

Show that

$$C = \frac{Mt^2 - 3t^2e^{-t}}{1+t+t^2}$$

[5]

(e) It is found that the long-term concentration is 10, and C reaches half this value after 105 minutes. Determine which of the models proposed in parts (c) and (d) is more consistent with this data. **[2]**